Artificial Intelligence & Machine Learning – Task 3

Model Validation, Overfitting Control & Hyperparameter Tuning

Objective

The objective of this project was to improve the House Price Prediction system developed in Task 2 by applying model validation techniques, detecting overfitting, performing hyperparameter tuning, and selecting the most reliable machine learning model. The California Housing Dataset was used to maintain consistency with the previous task.

Introduction

In machine learning, achieving high accuracy on training data alone is not sufficient. A model must also perform well on unseen data. Models that perform extremely well on training data but poorly on new data are said to be overfitting.

This project focuses on identifying overfitting, validating model performance using cross-validation, and optimizing model parameters using GridSearchCV. These techniques are commonly used in real-world machine learning projects to improve model reliability and generalization.

Dataset Description

The California Housing Dataset from Scikit-Learn was used in this project.

Target Variable

- HousePrice (Median House Value)

Input Features

- MedInc (Median Income)
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup

- Latitude
- Longitude

The dataset contains housing-related information used to predict median house prices.

Methodology

Step 1: Data Loading

The California Housing Dataset was loaded using Scikit-Learn and converted into a Pandas DataFrame for analysis.

Step 2: Feature Scaling

StandardScaler was applied to normalize feature values and ensure that all features contributed equally during model training.

Step 3: Train-Test Split

The dataset was divided into training and testing sets using an 80:20 ratio.

- Training Data: 80%
- Testing Data: 20%

This allowed model evaluation on unseen data.

Overfitting Analysis

An unrestricted Decision Tree Regressor was trained to identify overfitting behavior.

Results

Metric	Value
Training RMSE	0.0000
Testing RMSE	0.7030
Difference	0.7030

Interpretation

The training RMSE was almost zero, indicating that the model memorized the training data perfectly. However, the testing RMSE was significantly higher.

This large difference confirms that the model suffered from severe overfitting and could not generalize effectively to unseen data.

Cross-Validation

To obtain a more reliable estimate of model performance, 5-Fold Cross-Validation was applied.

Cross-Validation RMSE Scores

Fold	RMSE
1	0.8877
2	0.8278
3	0.8985
4	0.9451
5	0.9195

Average Cross-Validation RMSE

0.8957

Interpretation

Cross-validation evaluates the model on multiple train-test combinations rather than relying on a single split.

The average RMSE of 0.8957 provides a more realistic estimate of the model's performance and reduces the risk of biased evaluation.

Hyperparameter Tuning

GridSearchCV was used to identify the best Decision Tree parameters.

Parameter Grid

- max_depth = [3, 5, 7, 10]
- min_samples_split = [2, 5, 10]

Best Parameters

Parameter	Value
max_depth	10
min_samples_split	10

Best Cross-Validation RMSE

0.6366

Interpretation

The optimized parameters reduced model complexity and controlled overfitting while maintaining strong predictive performance.

Evaluation of Optimized Model

The tuned Decision Tree model was evaluated on the testing dataset.

Performance Metrics

Metric	Value
RMSE	0.6454
R^2 Score	0.6821

Interpretation

The optimized model achieved a lower prediction error and higher explanatory power compared to the baseline models.

Model Comparison

Model	RMSE	R^2 Score
Linear Regression	0.7456	0.5758
Ridge Regression	0.7456	0.5758
Tuned Decision Tree	0.6454	0.6821

Discussion

The comparison results indicate that the Tuned Decision Tree outperformed both Linear Regression and Ridge Regression.

Key observations:

1. The Tuned Decision Tree achieved the lowest RMSE.
2. It achieved the highest R^2 Score.
3. Hyperparameter tuning successfully reduced overfitting.
4. Cross-validation confirmed improved model reliability.

Although Linear Regression and Ridge Regression are simpler models, they were unable to capture complex relationships in the housing dataset as effectively as the tuned Decision Tree.

Final Model Selection Justification

The Tuned Decision Tree was selected as the final model because:

- It achieved the highest R^2 Score (0.6821).
- It achieved the lowest RMSE (0.6454).
- Hyperparameter tuning reduced overfitting.
- Cross-validation demonstrated stable performance.
- It provided better predictive accuracy than the baseline models.

Therefore, the Tuned Decision Tree represents the most reliable and effective solution for house price prediction in this project.

Conclusion

This project demonstrated the importance of model validation and hyperparameter tuning in machine learning. An initial Decision Tree model showed clear signs of overfitting, which was identified through train-test performance comparison.

Cross-validation provided a more reliable estimate of performance, while GridSearchCV optimized model parameters and improved generalization. The final Tuned Decision Tree model achieved the best overall performance with an RMSE of 0.6454 and an R^2 Score of 0.6821.

The project successfully achieved its objective of building a more reliable and production-ready machine learning model for house price prediction.